School of Computing and Creative Technologies

Assessment Specification – Portfolio Task 2

Module Details

Module Code	UFCE8B-30-3
Module Title	Data Science for Cyber Security
Module Leader	Ahsan Ikram
Module Tutors	Ahsan Ikram, Andrew McCarthy
Year	2025-26
Task	02
Total number of assessments for this module	01
Weighting	25%
Date issued to students	24/11/2025

Task Specification

You are a cyber security data analyst consulting law enforcement in trying to establish a timeline of events and track the spread of misinformation based on various datasets to create an attacker profile in the aftermath of an incident. One of the datasets you are analysing is a dump of social media posts around the time of the incident. You have collected data from various sources that shows content being shared on a social media platform around the incident. Some of the content appears to be targetted misinformation campaigns. You want to perform further diagnostic analysis on the datase.

Dataset: fake_news.csv (Available on Blackboard.)

Dataset description: The dataset provided is of a collection of fake and real news articles. The dataset has 20 columns.

1. *id*
2. *intext content*
3. *language*
4. *language of the article*
5. *crawled*
6. *crawled from where*
7. *site_url*
8. *site source*
9. *country*
10. *country name*
11. *domain_rank*
12. *thread_title*
13. *spam_score*
14. *main_img_url*
15. *replies_count*
16. *participants_count*
17. *likes*
18. *comments*
19. *shares*
20. *type*

You must complete this portfolio task using Python programming language.

Deliverable: Prepare a single Jupyter notebook. You might need skipping bad lines **(on_bad_lines='skip')** option added to the **'read_csv'** function.

Using the assigned dataset you have to perform the following tasks,

	Task	**Marks**
1	Prepare a correlation matrix for all numerical columns. Identify top three strongest correlated columns pairs. Print a summary.	3
2	Apply linear regression to the correlated columns identified in (1). Present the regression lines with clearly labelled visualisations.	6
3	Extract all the data labelled 'conspiracy', 'hate' and 'satire'. The labels are provided in the 'type' column. Train an appropriate Naïve Bayes classifier using the data (80-20 train-test split). Test the accuracy of the trained model with five samples from each type (conspiracy, hate and satire).	6
4	Using the 'author', 'shares', 'type' and 'title' columns, create a node link graph, using the data for 'hate', 'satire' and 'junksci' type data items. The graph should be created such that, - 'author' and 'title' are nodes, titles connected to the authors. - The shape/icon of the nodes should represent the 'type'. - The size/colour of the node represents the 'shares' value on a scale. The scale should be calculated observing the minimum and maximum shares. For example, 0-100, 100-500, 500-1000.	6
5	Using the 'text' column data for the rows labelled, 'bias' and 'conspiracy', - Prepare the data by removing stopwords, applying tokenization and lemmatization. - Prepare word clouds for both the categories ('bias' and 'conspiracy').	4
	Total	**25**